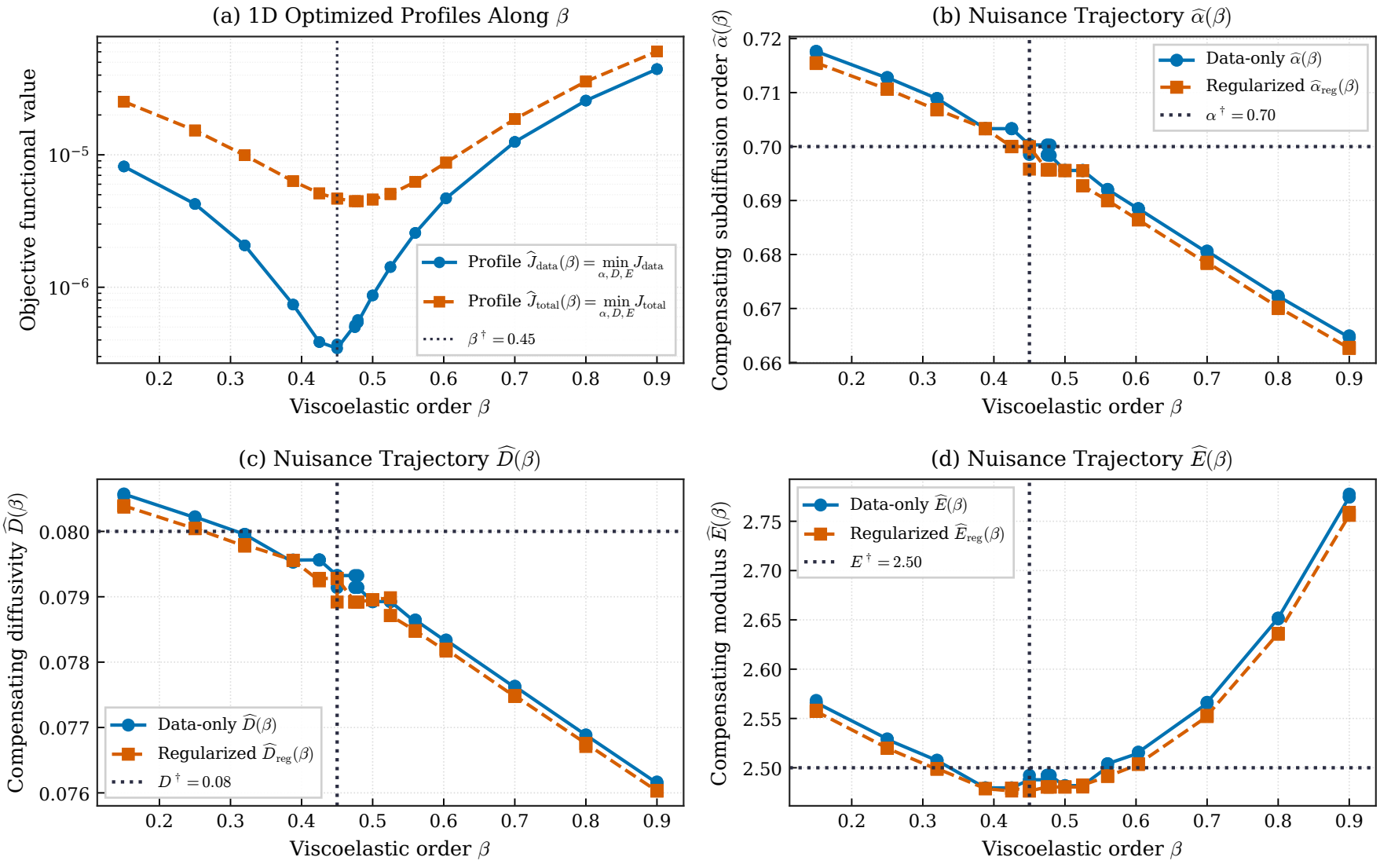
(a) 1D Optimized Profiles Along β
Objective functional value
10^{-5}
10^{-6}
0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9
Viscoelastic order β
Profile $\widehat{J}_{\text{data}}(\beta) = \min_{\alpha, D, E} J_{\text{data}}$
Profile $\widehat{J}_{\text{total}}(\beta) = \min_{\alpha, D, E} J_{\text{total}}$
$\beta^\dagger = 0.45$
(b) Nuisance Trajectory $\widehat{\alpha}(\beta)$
Compensating subdiffusion order $\widehat{\alpha}(\beta)$
0.72 0.71 0.70 0.69 0.68 0.67 0.66
0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9
Viscoelastic order β
Data-only $\widehat{\alpha}(\beta)$
Regularized $\widehat{\alpha}_{\text{reg}}(\beta)$
$\alpha^\dagger = 0.70$
(c) Nuisance Trajectory $\widehat{D}(\beta)$
Compensating diffusivity $\widehat{D}(\beta)$
0.080 0.079 0.078 0.077 0.076
0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9
Viscoelastic order β
Data-only $\widehat{D}(\beta)$
Regularized $\widehat{D}_{\text{reg}}(\beta)$
$D^\dagger = 0.08$
(d) Nuisance Trajectory $\widehat{E}(\beta)$
Compensating modulus $\widehat{E}(\beta)$
2.75 2.70 2.65 2.60 2.55 2.50
0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9
Viscoelastic order β
Data-only $\widehat{E}(\beta)$
Regularized $\widehat{E}_{\text{reg}}(\beta)$
$E^\dagger = 2.50$